

Automatic testing of web a11y

From zero level to dev focused



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Current state of accessibility in web development

- Accessibility topics are common on frontend conferences / meetups
- June 28th, 2025 - European Accessibility Act requires service compliance at WCAG 2.1 AA level ⁽¹⁾
- 88% of websites are not fully compliant with latest standards ⁽²⁾⁽³⁾
- Because we all are having excuses



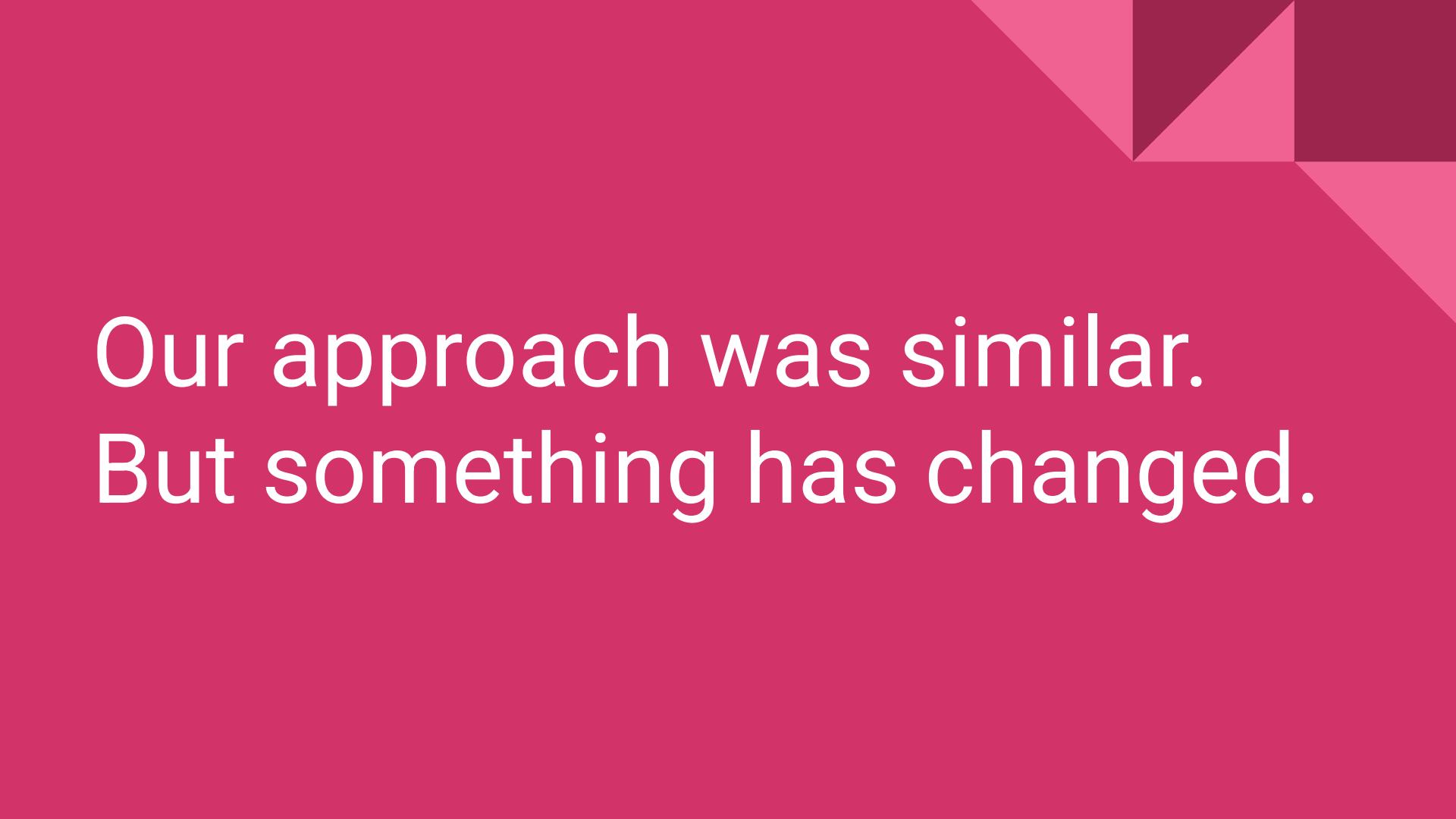
Olivia Richter

Web designer

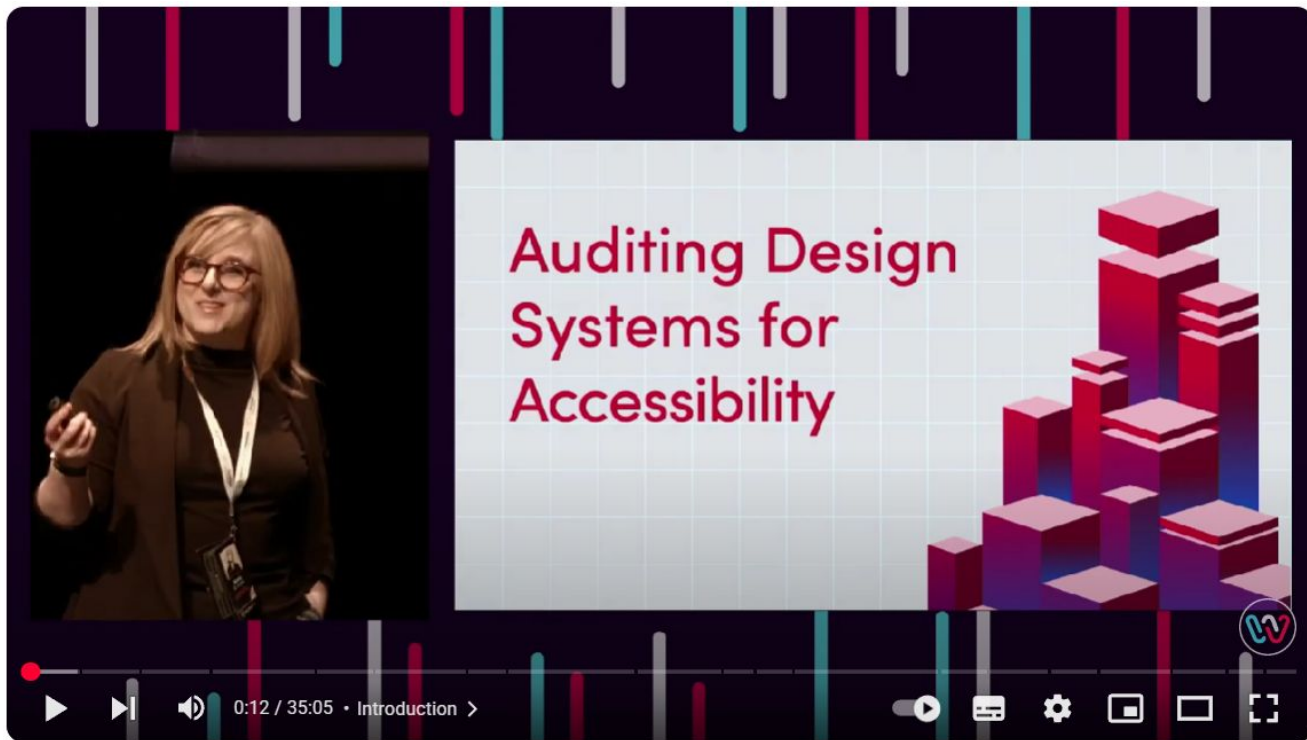
Accessible websites

Life-long learning





Our approach was similar.
But something has changed.



<https://www.youtube.com/watch?v=L0PRzlyP1Zc>

Auditing Design Systems for Accessibility by Anna E. Cook



“Accessibility is not about being perfect. It’s about trying.”

Anna E. Cook, Wey Wey Web 2023



“It’s best to try a11y testing on new applications. You start small and grow as the project grows.”

Source unknown

How to approach automatic testing by devs?

- Semantic and well described HTML is the key
- No need to test using all assistive technologies
- It's all about HTML - we should test production bundle
- Playwright to be the tool of our choice
(however, it's not obligatory)



Example repository



Add airport

Name

IATA Code

Country

Regions

- ☐ Aegean Islands
- ☐ Corfu Island (Kerkyra)
- ☐ Ha Long Island
- ☐ Rhodes Island

- ☒ Balearic Islands
- ☐ Crete Island
- ☐ Ionian Islands
- ☐ Tenerife Island

- ☐ Canary Islands
- ☐ Dodecanese Islands
- ☒ Maiorca Island

Vaccination Notes

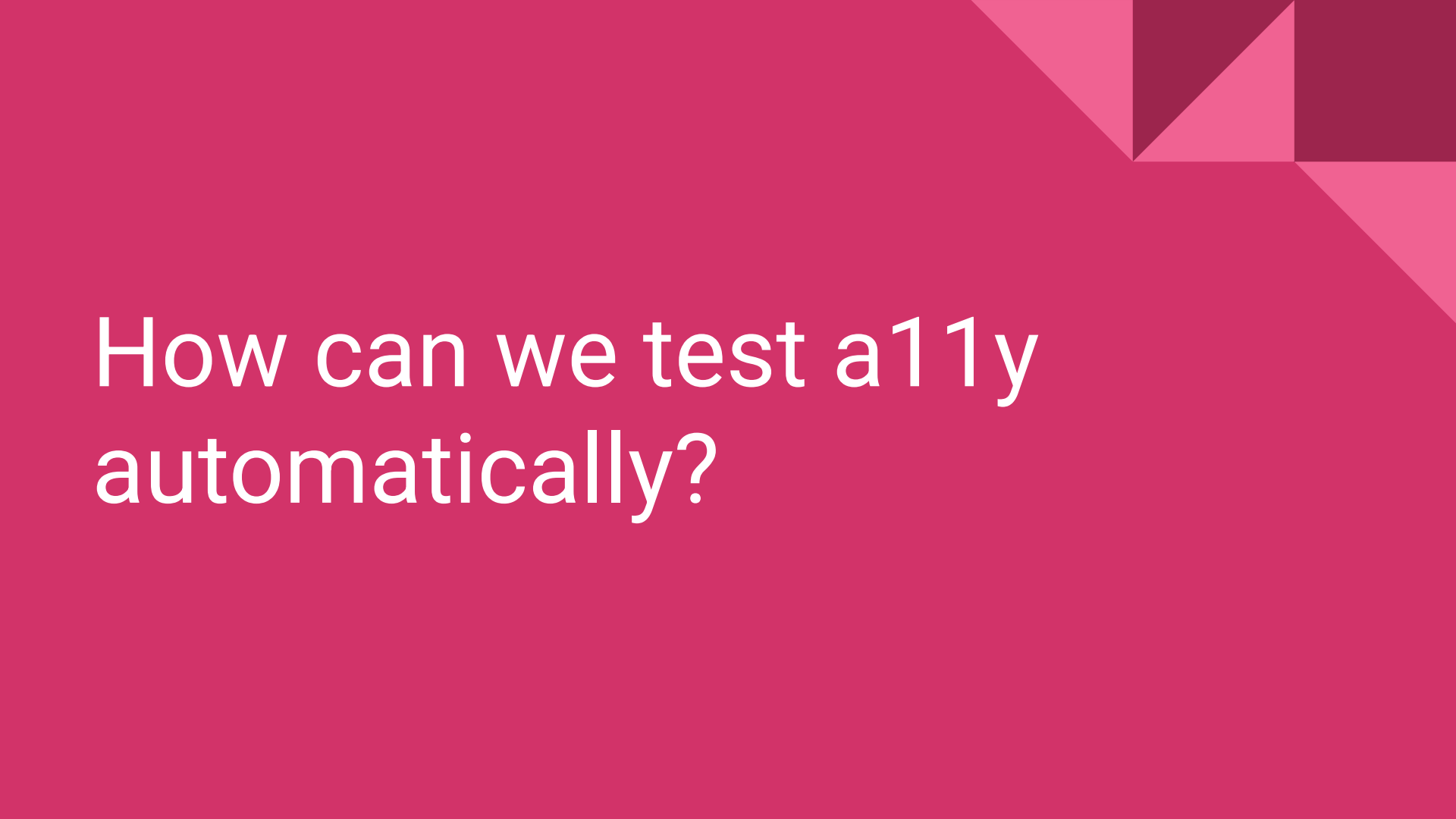
Example repository

- Fully functional basis of React SPA + Backend
- Reliable test examples
- Framework and reusables to run tests
- Note: The app is not a11y compliant!
(But you will find it out on test report)



<https://github.com/LukaszNowakPL/web-a11y-testing>





How can we test a11y
automatically?

User journey with keyboard navigation

- It's just another functional test - a low hanging fruit
- Minimal approach: cover at least a simplest happy path user journey



Test case actions

```
// Entering the page
public async proceedThroughPage(airport: AirportModel) {
    await this.assertNoFocus();
    await this.navigateToNextElement(); // ...

    // Filling airport name field
    await this.assertFocusedElement(this.page.getByRole('textbox', {name: /name/i}));
    await this.fill(airport.name);
    await this.navigateToNextElement(); // ...

    // Clicking submit button
    await this.assertFocusedElement(this.page.getByRole('button', {name: /send/i}));
    await this.performAction();
}
// Asserting data sent to backend service
```

Reusable helpers

```
private getFocusedElement () {  
    return this.page.locator('*:focus') as NotClickableLocator;  
}  
  
public async assertFocusedElement (expectedElement: Locator) {  
    expect(await this.getOuterHtml(this.getFocusedElement())).toStrictEqual(await this.getOuterHtml(expectedElement));  
}  
  
private async getOuterHtml (element: Locator) {  
    return await element.evaluate ((node) => node.outerHTML);  
}  
  
public async performAction () {  
    await this.page.press('body', this.KEYS.action);  
}  
  
public async selectFocusedElement () {  
    await this.page.press('body', this.KEYS.selection);  
}  
  
public async navigateToNextElement () {  
    await this.page.press('body', this.KEYS.navigateToNextElement);  
}
```


Static HTML audits

Same page, different states

Provide piece of data

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Piece of data

Submit

Provide piece of data

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Piece of data

This data is obligatory

Submit

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Piece of data

Data required

Submit

Test case actions

```
// Then I wait until the page is ready  
await addAirportPage.assertReady();  
  
// When I perform snapshot analysis for initial view  
await addAirportPage.performSnapshotAnalysis('Initial view', {takeLighthouseSnapshot});  
  
// And I fulfill the form with correct data  
await addAirportPage.fulfillForm(airport);  
  
// Then I can see the form data is valid and Submit button is enabled  
await addAirportPage.assertSubmitEnabled();  
  
// When I perform snapshot analysis for provided data  
await addAirportPage.performSnapshotAnalysis('Form filled', {takeLighthouseSnapshot});  
// ...
```

The audit process

```
public async assertSnapshots(snapshotTitle: string, config?: AssertSnapshotConfig) {
  await Promise.all([
    this.assertAxeChecks(snapshotTitle, Boolean(config?.skipAxeViolationChecks)),
    this.performAriaSnapshotComparison(snapshotTitle),
    config?.takeLighthouseSnapshot !== undefined
      && this.assertLighthouseSnapshot (snapshotTitle,
config.takeLighthouseSnapshot ),
  ]);
}
```

Lighthouse audit

- playwright-lighthouse for integration
- Accessibility audit is nicely covered
 - Best practices, WCAG checks, HTML semantics
 - Visual analysis (colour contrasts, distances, etc.)
- Provides helpful, human readable report

Sausage fingers
mentioned by
Caitlin



<https://www.npmjs.com/package/playwright-lighthouse>



Using Lighthouse Flow

```
// And Lighthouse flow
const flow = await LighthousePlaywrightFlow
    .startFlow('Airport addition journey', page);

// And take snapshot helper to be passed to each page
const takeLighthouseSnapshot: TakeLighthouseSnapshot = async (name) =>
    await flow.snapshot(name,
        {onlyCategories: Object.keys(lighthouseThresholds), formFactor: 'mobile'});

// ...
// When I perform snapshot analysis for initial view
await addAirportPage.performSnapshotAnalysis('Initial view',
    {takeLighthouseSnapshot, skipAxeViolationChecks: true});

// ...
// Then I generate snapshot report
await flow.generateReports(testInfo, 'lighthouse-test-add-airport-happy-path-flow.html',
    lighthouseThresholds);
```

Creating Lighthouse flow report

```
const res = await this.flow.createFlowResult ();

// Attach Lighthouse report to Playwright report
await testInfo.attach(filename, {body: reports.toString()});

const lighthouseThresholds = {'best-practices': 100, accessibility: 90, seo: 50};

const categories = Object.keys(lighthouseThresholds)
  as ('accessibility' | 'best-practices' | 'seo')[];
let thresholdFailures: string[] = [];

res.steps.forEach((step) => { /* ... */ });

expect(thresholdFailures).toStrictEqual([]);
```

Axe accessibility testing engine

- Static code analysis provided by industry leader
- Produces raw, textual, in-depth reports
- Restricts scanning against certain criterias (i.e. up to WCAG 2.1 AA)



<https://playwright.dev/docs/accessibility-testing>



Using Axe static code analysis

```
private async assertAxeChecks (snapshotTitle: string, skipViolationChecks: boolean) {
    const axeChecks = await new AxeBuilder({page: this.page})
        .withTags(['wcag2a', 'wcag2aa', 'wcag21a', 'wcag21aa', 'wcag22aa'])
        .analyze();

    await this.testInfo.attach(this.composeAlllyReportName (snapshotTitle), {
        body: JSON.stringify(axeChecks, null, 2),
        contentType: 'application/json',
    });

    if (!skipViolationChecks) {
        expect(axeChecks.violations).toEqual([]);
    }
}
```

Aria snapshot

- Accessibility tree idea ⁽⁵⁾
- It is like a DOM dedicated for a11y
- Playwright Aria snapshots functionality
- Accessibility regression pattern



<https://playwright.dev/docs/aria-snapshots#aria-snapshots>



Aria snapshot regression

```
private async performAriaSnapshotComparison (snapshotTitle: string) {
  expect(await this.page.locator('body').ariaSnapshot()).toMatchSnapshot([
    ARIA_SNAPSHOTS_DIRECTORY,
    `${this.composeSnapshotName(snapshotTitle)}.yaml`,
  ]);
}

private composeSnapshotName (snapshotTitle: string) {
  return slugify(`${this.pageTitle} ${this.testSuite} ${snapshotTitle}`);
}
```

Aria snapshot example

- `banner`:
 - `link "Airports dashboard"`:
 - `img`
 - `text`: Airports dashboard
 - `navigation`:
 - `list`:
 - `listitem`:
 - `link "Airports"`
 - `listitem`:
 - `link "Add airport"`
 - `main`:
 - `alert`: Sorry, there is some connectivity error
 - `paragraph`:
 - `text`: We were unable to fetch some of data necessary to display the page. You can
 - `button "restart data fetching"`
 - `text`: manually.
 - `paragraph`: If it won't help, try to restart the app.

Aria snapshot example



Airports dashboard

Airports **Add airport**

Sorry, there is some connectivity error

We were unable to fetch some of data necessary to display the page. You can [restart data fetching](#) manually.
If it won't help, try to restart the app.

Unknown super-power of aria snapshot

```
- listitem:  
  - link "Airports"  
- listitem:  
  - link "Add airport"  
- main:  
  - heading "Add airport" [level=1]  
  - text: Name  
  - textbox "Name"  
  - text: IATA Code  
  - textbox "IATA Code"  
  - text: Country  
  - combobox "Country": Select one  
  - text: Regions
```

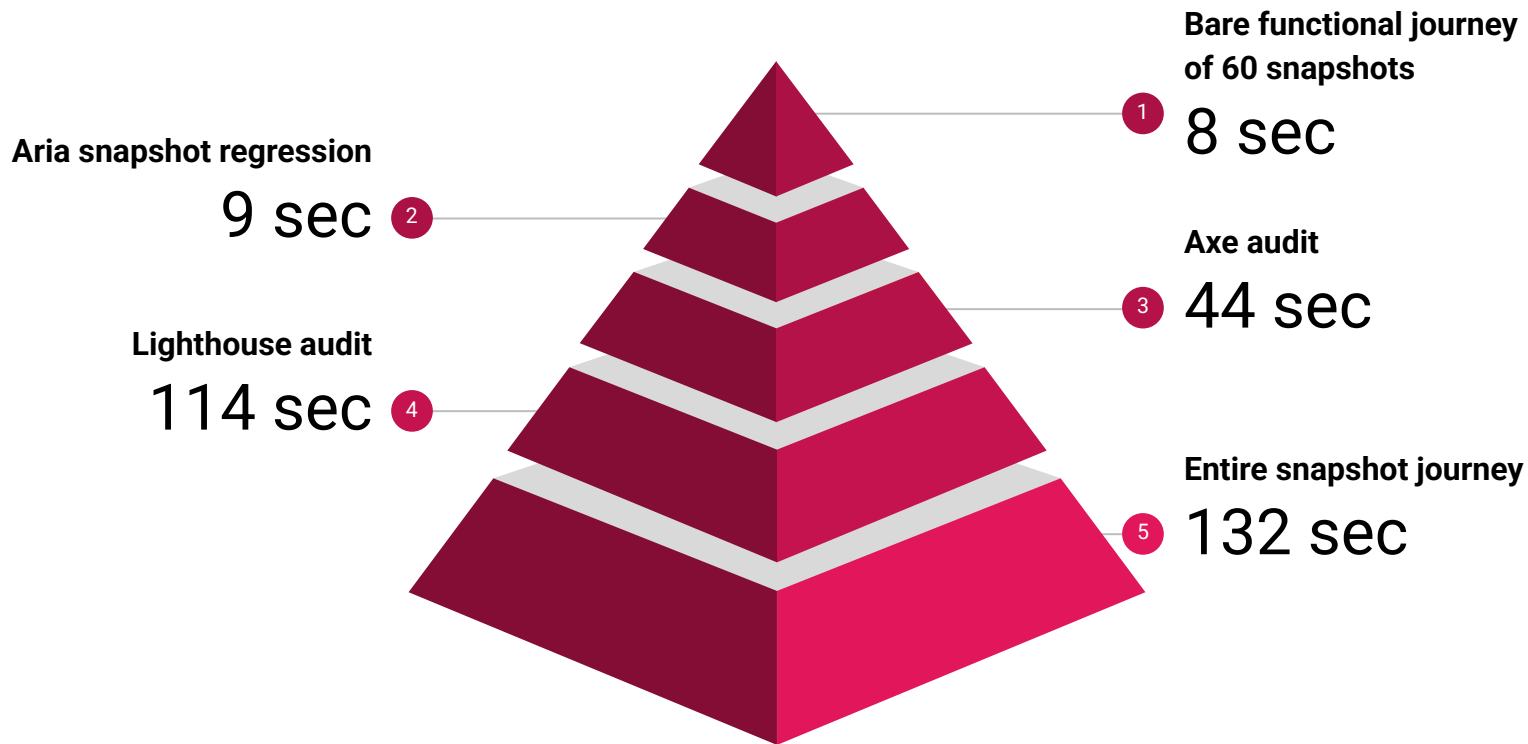
Gotchas and findings

Gotchas and findings

- Snapshot suite is time consuming



Snapshot execution times



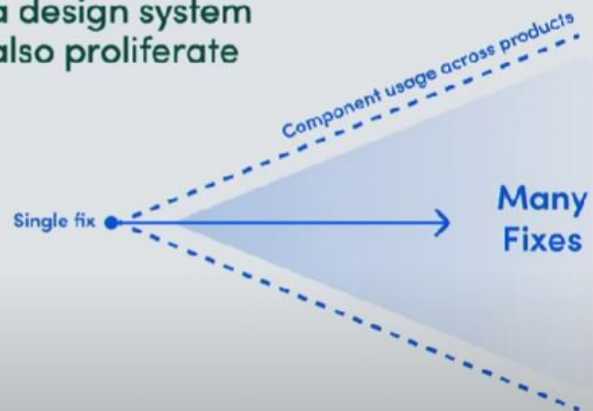
Gotchas and findings

- Snapshot suite is time consuming
- Snapshot idea helps to understand the view lifecycle
- It's crucial to cover a11y on design system





Accessibility fixes in
a design system
also proliferate



16



8:39 / 35:05 • Components >



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Gotchas and findings

- Snapshot suite is time consuming
- Snapshot idea helps to understand the view lifecycle
- It's crucial to cover a11y on design system
- Devs to test their own application, while QAs to focus on entire product
- A11y testing is relatively new thing. It lacks global solutions, toolings or good practice definitions



Same page, different states

Provide piece of data

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Piece of data

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- Snapshot idea helps to understand the view lifecycle
- It's crucial to cover a11y on design system
- Devs to test their own application, while QAs to focus on entire product
- A11y testing is relatively new thing. It lacks global solutions, toolings or good practice definitions
- Beware of paid solutions



Conclusions

Conclusions

- A11y is not about being perfect. It's about trying.
- So, we did not become an a11y experts. We are, however, an a11y-aware.
- We are able to detect only small factor of a11y errors
- Tests with real users and assistive technologies is the key



Playwright disclaimer

DISCLAIMER

Automated accessibility tests can detect some common accessibility problems such as missing or invalid properties. But many accessibility problems can only be discovered through manual testing. We recommend using a combination of automated testing, manual accessibility assessments, and inclusive user testing.

For manual assessments, we recommend [Accessibility Insights for Web](#), a free and open source dev tool that walks you through assessing a website for [WCAG 2.1 AA](#) coverage.

Source: <https://playwright.dev/docs/accessibility-testing>



Q & A

References

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